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Illuminating coating for an element of a suspension assembly

Abstract

The suspension assembly may include a suspension element operably coupling a chassis to a wheel assembly of the vehicle, and a coating operably coupled to an exposed portion of the suspension element that is compressed or displaced responsive to a suspension event. The coating when powered may emit light to illuminate the suspension element and an area proximate to the suspension element.

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Background/Summary

TECHNICAL FIELD

(1) Example embodiments generally relate to suspension assembly components and, more particularly, relate to a coating for a suspension assembly element.

BACKGROUND

(2) The suspension assembly is an important aspect of a vehicle to an operator. As a portion the suspension assembly is exposed, an illuminated portion of the suspension assembly may visually highlight a specific suspension element. Standard illumination via LEDs adds increased complexity to the suspension assembly. As such, an illuminated coating without LEDs that may be applied to a suspension element may be desired.

BRIEF SUMMARY OF SOME EXAMPLES

(3) In accordance with an example embodiment, a suspension assembly of a vehicle may be provided. The suspension assembly may include a suspension element operably coupling a chassis to a wheel assembly of the vehicle, and a coating operably coupled to an exposed portion of the suspension element that is compressed or displaced responsive to a suspension event. The coating, when powered, may emit light to illuminate the suspension element and an area proximate to the

suspension element.

(4) In another example embodiment, a suspension element for operably coupling a chassis to a wheel assembly of a vehicle may be provided. The suspension element may include a first interface portion in contact with a chassis component, a second interface portion in contact with a wheel assembly component, a reaction portion disposed between the first interface portion and the second interface portion to be displaced or compressed responsive to a suspension event, an exposed portion externally visible when the suspension event occurs, and a coating operably coupled to the exposed portion of the suspension element that is displaced or compressed responsive to the suspension event. The coating, when powered, may emit light to illuminate the suspension element and an area proximate to the suspension element.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

(1) Having thus described the invention in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

(2) FIG. 1 depicts a block diagram of a suspension assembly for a vehicle in accordance with an example embodiment;

(3) FIG. 2, illustrates a perspective view of a suspension assembly in accordance with an example embodiment;

(4) FIG. 3 depicts a cross-section of a suspension assembly in accordance with an example embodiment; and

(5) FIG. 4, including FIGS. 4A and 4B, illustrates perspective views of a suspension element in accordance with an example embodiment.

DETAILED DESCRIPTION

(6) Some example embodiments now will be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all example embodiments are shown. Indeed, the examples described and pictured herein should not be construed as being limiting as to the scope, applicability or configuration of the present disclosure. Rather, these example embodiments are provided so that this disclosure will satisfy applicable requirements. Like reference numerals refer to like elements throughout. Furthermore, as used herein, the term “or” is to be interpreted as a logical operator that results in true whenever one or more of its operands are true. As used herein, operable coupling should be understood to relate to direct or indirect connection that, in either case, enables functional interconnection of components that are operably coupled to each other.

(7) Some example embodiments described herein may address the issues described above. In this regard, for example, some embodiments may provide a coating for a suspension element of a vehicle. As a result, the addition of the coating may provide illumination to the suspension element.

(8) FIG. 1 illustrates a block diagram of a suspension assembly **100** for a vehicle **110** in accordance with an example embodiment. As seen in FIG. 1, some cases, the vehicle **110** may include a chassis **120**. In an example embodiment, the chassis **120** may be a frame or body of the vehicle **110**. In some cases, the chassis **120** or frame may support and/or may form the foundation structure of the vehicle **110**. In an example embodiment, the chassis **120** and/or frame may be formed of one or more casted subframes, and a suspension element **200** may be operably coupled to the chassis **120** or frame to help operably couple a wheel assembly **130** to the chassis **120** or frame.

(9) In some cases, the suspension element **200** may be a variety of different components within the suspension assembly **100** of the vehicle **110**. In an example embodiment, the suspension element **200** may be a spring. For example, the suspension element **200** may be a coil spring, a pneumatic spring, a torsion spring, or any number of different types of springs (i.e. leaf spring) that may be present within the suspension assembly **100**. In some cases, the coil spring and/or the pneumatic

spring may be a component of a suspension damper for the suspension assembly **100**. The suspension damper may function to absorb compression and rebound loading along a longitudinal axis of the suspension damper. In this regard, the suspension damper may significantly limit oscillations and vibrations of the vehicle **110** by dampening the articulation motion of the wheel assembly **130** so that the articulation of the wheel assembly **130** is not directly transferred to the chassis **120** of the vehicle **110**.

(10) In an example embodiment, the suspension damper may be a vertical damper that, in some other cases, may be referred to as a shock absorber. In some cases, the suspension damper may include both a coil spring and a pneumatic spring. For example, a coil spring may surround a pneumatic spring to assist in vehicle damping. In an example embodiment, as the vehicle traverses uneven terrain and the wheel assembly **130** articulates towards the chassis **120**, the suspension damper may compress due to a compression loading force. In some cases, the suspension damper may be a monotube shock absorber, a twin-tube shock absorber, or a bladder shock absorber depending on the type of vehicle **110** or the desired damping control. In an example embodiment, the suspension damper may be active, semi-active, or passive. In an example embodiment, the active or semi-active damper may utilize measurements from a sensor suite of the vehicle **110** to adjust the suspension damper accordingly.

(11) In some cases, the suspension element **200** may be a torsion spring. In an example embodiment, the torsion spring may be a torsion bar and part of a torsion spring suspension assembly or torsion bar suspension assembly. In some cases, the suspension element **200** may be other elements within the suspension assembly **100**, including but not limited to a stabilizer bar, control arm, steering hub, suspension sleeve, suspension bellows, suspension cover or a suspension bushing. The suspension sleeve, suspension bellow, and suspension cover may be disposed over a coil spring or be an element of the suspension damper. In an example embodiment, the suspension element **200** may be multiple suspension elements. For example, there may be multiple coil springs or suspension dampers within the suspension assembly **100**.

(12) In some cases, the suspension element **200** may be operably coupled to a control module **140**. In an example embodiment, the control module **140** may be a controller. In some cases, the control module **140** may include one or more control modules (i.e. sub-control modules or operably coupled to other control modules). The control module **140** may include processing circuitry that includes a processor and memory. The processing circuitry may be configured to provide electronic control of the inputs to one or more functional units of the suspension assembly **100** and to process data received at or generated by the one or more functional units of the suspension assembly **100**. Thus, the processing circuitry may be configured to perform data processing, control function execution and/or other processing and management services according to an example embodiment. In some embodiments, the processing circuitry may be embodied as a chip or chip set. In other words, the processing circuitry may comprise one or more physical packages (e.g., chips) including materials, components and/or wires on a structural assembly (e.g., a baseboard). The structural assembly may provide physical strength, conservation of size, and/or limitation of electrical interaction for component circuitry included thereon. The processing circuitry may therefore, in some cases, be configured to implement an embodiment of the present invention on a single chip or as a single “system on a chip.” As such, in some cases, a chip or chipset may constitute means for performing one or more operations for providing the functionalities described herein. In an example embodiment, other vehicle control modules may include similar processing circuitry.

(13) The control module **140** may be operably coupled to the suspension element **200** and other vehicle components via a variety of methods. The control module **140** may utilize wired or wireless communications to communicate and receive information from vehicle components. In some cases, the control module **140** may receive information from other vehicle control modules connected to the vehicle **110** or external control modules (i.e. databases, service centers, subscription providers, etc.).

(14) In an example embodiment, the suspension element **200** may be operably coupled to a coating **150**. The coating **150** being operably coupled to the suspension element **200** may mean the coating **150** may be applied to the suspension element **200**. The coating **150** may operably couple to any number of portions of the suspension element **200**. In some cases, the coating **150** may operably couple to other components of the suspension assembly **100** and/or the vehicle **110**. In an example embodiment, the coating **150** may be a single-layered or multi-layered. In some cases, the coating **150** may provide corrosion resistance for the suspension element **200** and/or the suspension assembly **100**. The coating **150** may be a polymer-based, and the coating **150** may have elasticity that allows for movement of the suspension element **200** and/or the suspension assembly **100** without interrupting coverage of the coating **150**. The coating **150** may be applied to other elements of the vehicle **110** as well, such as but not limited to a wheel hub.

(15) In some cases, the suspension element **200** and the coating **150** may be operably coupled to integrated circuitry **160** and a sensor **170**. In an example embodiment, the integrated circuitry **160** and the sensor **170** may be embedded within the coating **150**. In an example embodiment, the integrated circuitry **160** may include the sensor **170**. In some cases, the sensor **170** may be a sensor suite and may include multiple sensors. In an example embodiment, the sensor **170** may provide active suspension telemetry data in real-time to the control module **140** of the vehicle **110**. The sensor **170** may be a strain gauge, a displacement sensor, a position sensor, a thermal sensor, a torque sensor, or any other sensor that may provide suspension data, vehicle data, or environment data to the vehicle **110**. The control module **140** may utilize the active suspension telemetry data to adjust vehicle modes or the suspension assembly **100** depending on the specific data received. The active suspension telemetry data may only be a single input of data of a larger data pool that the control module **140** may utilize to make adjustments to the vehicle **110**.

(16) FIG. 2 illustrates a perspective view of a suspension assembly in accordance with an example embodiment. FIG. 3 depicts a cross-section of a suspension assembly in accordance with an example embodiment. FIG. 4 illustrates perspective views of a suspension element in accordance with an example embodiment. In some cases, the suspension element **200** may include a first interface portion **201** and a second interface portion **202**. A first interface portion **201** may be in contact with the chassis **120** and/or may operably couple the suspension element **200** to the chassis **120**. The second interface portion **202** may be in contact with the wheel assembly **130** and/or may operably couple the suspension element **200** to the wheel assembly. For example, if the suspension element **200** is a coil spring, a first distal end of the coil spring may be the first interface portion **201** to operably couple the chassis **120** and the suspension element **200**, and a second distal end of the coil spring may be the second interface portion **202** to operably couple the suspension element **200** and the wheel assembly **130**.

(17) In a further example, if the suspension element **200** is a suspension bushing, an exterior surface of the suspension bushing may be the first interface portion **201**, and an interior surface of the suspension bushing may be the second interface portion **202**. In some cases, the first interface portion **201** may be the internal surface of the suspension bushing, and the second interface portion **202** may be the external surface of the suspension bushing. In an example embodiment, the first interface portion **201** and the second interface portion **202** may operably couple with other suspension assembly components to operably couple the suspension element **200** with the chassis **120** and the wheel assembly **130** respectively. In some cases, the other suspension assembly components may include a suspension bushing, a spring rest/stop, or any other connection/attachment mechanisms that may operably couple the suspension element **200** with the chassis **120** and the wheel assembly **130**.

(18) In an example embodiment, the suspension element **200** may include a reaction portion **203**. The reaction portion **203** may be disposed between the first interface portion **201** and the second interface portion **202**. In some cases, the reaction portion **203** may not have to be directly disposed between the first interface portion **201** and the second interface portion **202**. For example, the

reaction portion **203** may be disposed proximate the first interface portion **201** and the second interface portion **202**. The reaction portion **203** may be displaced or compressed responsive to a suspension event. A suspension event may be any event that may cause displacement or compression with the suspension assembly **100**. For example, the suspension event may be caused by the vehicle **110** experiencing a rough driving surface or while the vehicle **110** is off-roading. For example, as seen in FIG. 4, if the suspension element **200** is a spring, the suspension element **200** may be an undisplaced spring **401** prior to a suspension event in FIG. 4A. Responsive to the suspension event, the undisplaced spring **401** may be compressed to be positioned as the displaced spring **402** in FIG. 4B. The reaction portion **203** may be the primary portion of the spring changed between the undisplaced spring **401** and the displaced spring **402** responsive to the suspension event occurring. In an example embodiment, the suspension event may be caused by the vehicle **110** via the control module **140**.

(19) In some cases, the suspension element **200** may include an exposed portion **204**. The exposed portion **204** of the suspension element **200** may be visible from the exterior of the vehicle **110** when the suspension event occurs. In an example embodiment, the exposed portion **204** may be a percentage of, a section of, or may overlap with the reaction portion **203** of suspension element **200**. In some cases, the exposed portion **204** and/or the reaction portion **203** of the suspension element **200** may be a percentage of, a section of, or may overlap with the first interface portion **201** and the second interface portion **202** of suspension element **200**.

(20) In some cases, the coating **150** may be operably coupled to the reaction portion **203** of the suspension element **200**. In an example embodiment, the coating **150** may be operably coupled to the exposed portion **204** of the suspension element **200**. In some cases, the coating **150** may be operably coupled to a section of the suspension element **200** that includes both the reaction portion **203** and the exposed portion **204**. The coating **150** may be operably coupled to a variety of portions of the suspension element **200**, including but not limited to the first interface portion **201**, the second interface portion **202**, the reaction portion **203**, and the exposed portion **204**.

(21) In an example embodiment, the coating **150** may emit light. The coating **150** may emit light when powered to illuminate the suspension element **200** and/or an area proximate to the suspension element **200** (i.e. the suspension assembly **100** or other vehicle components/environment). The coating **150** may emit light based on an electroluminescent property of the coating. In some cases, a variety of different methods and materials may provide the electroluminescent property to the coating **150**. For example, the electroluminescent property may be added to the coating **150** via zinc sulfide compositions with different additional metal types. For instance, depending on the metal type (i.e. copper, silver, manganese, etc.) added with the zinc sulfide compositions, the color and other properties of the electroluminescent property of the coating **150** may vary.

(22) In an example embodiment, the coating **150** may only emit light when powered. In some cases, the coating **150** may be powered by a wired connection to one or more onboard power sources of the vehicle **110**. In some cases, the one or more onboard power sources of the vehicle **110** may be a primary battery or other battery of the vehicle **110**. In an example embodiment, the one or more onboard power sources may be primary power sources for other vehicle components, including but not limited to the control module **140**, vehicle sensor suites, and other powered suspension assembly/vehicle components. In some cases, the one or more onboard power sources may be integrated within the suspension element **200**. For example, the suspension element **200** may include its own battery or power source.

(23) In some cases, the coating **150** may be powered via a piezoelectric element. The piezoelectric element may utilize movement or displacement of the suspension element **200** and convert the movement or displacement of the suspension element to electrical energy. In an example embodiment, the movement or displacement of the suspension element **200** may be a compression, relaxation, and/or displacement of a spring at the reaction portion **203** of the suspension element **200** and may be converted into electrical power via the piezoelectric element.

(24) In some cases, the coating **150** may be powered via a thermoelectric element. The thermoelectric element may utilize heat created via movement or displacement of the suspension element **200** and convert the heat to electrical energy. In an example embodiment, the movement or displacement of the suspension element **200** may be a compression of a spring at the reaction portion **203** of the suspension element **200**, and the heat created from the compression of the suspension element **200** may be converted into electrical power via the thermoelectric element. In some cases, the coating **150** may be powered via a radio frequency (RF) power element. The RF power element may be a wireless power transfer element that picks up low-level radio frequency waves from a source and converts the wave's energy to electrical power.

(25) In an example embodiment, the one or more onboard power sources, the piezoelectric element, and/or the thermoelectric element may provide electrical power directly to the coating **150** to utilize/activate its electroluminescent property. In this regard, the coating **150** may emit light without the need for an explicit lighting elements or LEDs. The lack of explicit lighting elements or LEDs may allow for less complexity in the construction, assembly, or instillation of the suspension assembly **100**. In some cases, the one or more onboard power sources, the piezoelectric element, and/or the thermoelectric element may be included with and/or in the integrated circuitry **160**. In an example embodiment, the one or more onboard power sources, the piezoelectric element, and/or the thermoelectric element may be embedded within the coating **150**.

(26) The integrated circuitry **160** may include wires, one or more electrical connections, sensor circuitry, sensor hardware, and any other circuitry types that may provide additional functionality to the suspension element **200** and the suspension assembly **100**. In some cases, the one or more electrical connections of the suspension element **200** may operably couple to other components of the vehicle **110** (i.e. the control module **140**). In an example embodiment, the one or more electrical connections may be disposed at a substantially stationary portion of the suspension element **200**. Substantially stationary may be considered stationary relative to other portions of the suspension element **200** or less than 10 cm of displacement of the portion from its initial position. The substantially stationary portion of the suspension element **200** may be the first interface portion **201** and/or the second interface portion **202**. Positioning the one or more electrical connection at the substantially stationary portion of the suspension element **200** may ensure secure performance of the one or more electrical connections, as the limited movement at the substantially stationary portion may prevent incidental disconnection of the one or more electrical connections.

(27) In some cases, conductive ink printing may form the integrated circuitry **160** of the suspension element **200**. Conductive ink printing may use a ink jet printer to apply very precise circuits to a variety of objects, including fabrics and rigid objects. The conductive ink may be a liquid-based ink. With conductive ink printing, a computer-aided design (CAD) of circuitry or electronics may be directly incorporated during or after the primary manufacturing or assembly process of the suspension element **200** or suspension assembly **100** without losing specificity. Conductive ink printing may include conductive inks composed with conductive metal fillers and polymer resins. The conductive metal fillers may provide electrical and thermal properties. For example, the conductive metal filler may be silver, as the conductive ink needs to be easily liquefied at relatively low temperatures (i.e. 500° F. or less), while maintaining its conductivity through the printing process and application. In some cases, the temperature threshold may be higher (i.e. 2000° F. or less) depending on the application. The polymer resins may provide mechanical strength and flexibility in application of the conductive ink printing. For example, silver nitrate may be a commonly used ink for conductive ink printing. Forming the integrated circuitry **160** via conductive ink printing may allow for increased flexibility of the suspension element **200** and suspension assembly **100**. Additionally, the conductive ink printing may provide increased ease of integration of the integrated circuitry **160**. Conductive ink printing may also be conductive ink painting.

(28) In some cases, the sensor **170** may be embedded within the coating **150** via conductive ink

printing. For example, if the sensor **170** is a strain gauge, specific strain gauge patterns may be integrated within the coating **150**. Conductive ink printing may ease the integration of specific and/or complex strain gauge patterns within the suspension element **200**. The locations of the sensor **170** within the suspension element **200** may vary, as well. For instance, the sensor **170** may be disposed at a location of the suspension element **200** that may experience the most displacement. For example, if the suspension element **200** is a coil spring of a vertical damper, the sensor **170** may be disposed at the distal end of the coil spring closest to the wheel assembly **130**. In some cases, the location may be the reaction portion **203** of the suspension element **200** to provide real-time data of the compression and/or displacement of the suspension element **200**. In an example embodiment, the sensor **170** may be disposed at a location of the suspension element **200** that provides specific data of the suspension assembly **100**. For example, in some cases if the suspension element **200** is a coil spring for a vertical damper, the sensor **170** may provide spring displacement data for end stop control of the suspension element **200**. In an example embodiment, the sensor **170** may be disposed at any portion of the suspension element **200**, including the first interface portion **201**, the second interface portion **202**, the reaction portion **203**, and/or the exposed portion **204**.

(29) In some cases, the sensor **170** may detect the presence of the suspension event and/or be used to classify the type of suspension event. In an example embodiment, the suspension event may be classified via the amount of displacement or compression of the suspension element **200**. In some cases, responsive to the suspension event, an intensity of the light emitted from the suspension element **200** may vary. For example, if a suspension event is not occurring, the suspension element **200** may not emit light. However, if a suspension event is occurring, responsive to the suspension event, the suspension element **200** may emit light.

(30) In some cases, the intensity of light emitted from the suspension element **200** may vary proportionally with the suspension event. For example, if the suspension element **200** is a spring and the suspension event is the compression of the spring, the intensity of the light emitted may increase as the compression of the spring increases. In the same regard, the intensity of the light emitted may decrease as compression of the spring decreases. In an example embodiment, the intensity of light emitted from the suspension element **200** may be inversely proportional with the suspension event.

(31) In some cases, the intensity of the light emitted from the suspension element **200** may vary based upon other events and/or instructions other than suspension events. In an example embodiment, the intensity of the light emitted from the suspension element **200** may vary based on the time of day or the amount of sunlight/light around the vehicle **110**. In some cases, the intensity of the light emitted from the suspension element **200** may vary based on inputs from the control module **140**. The inputs from the control module **140** may be from an operator of the vehicle or automatic based on data from a sensor **170** of the vehicle **110**.

(32) In an example embodiment, if the suspension element **200** is a spring, the coating **150** may adjust a multi-rate spring characteristic of the spring. In some cases, the multi-rate spring characteristic may be a change in spring rate as the spring is displaced. Spring rate may be a measurement of the amount of force needed to compress a spring a certain distance. For example, as seen in FIG. 4, the suspension element **200** may be a spring. The undisplaced spring **401**, as seen in FIG. 4A, may be the suspension element **200** prior to a suspension event. The undisplaced spring **401** may have a first spring rate. Responsive to a suspension event, the undisplaced spring **401** may be displaced to the state represented by the displaced spring **402**, as seen in FIG. 4B. The displaced spring **402** may have a second spring rate. In some cases, the first spring rate may be greater than the second spring rate (i.e. spring rate decreases as the spring is displaced/compressed). In an example embodiment, the difference between the first spring rate and the second spring rate may be proportional to the suspension event (i.e. the amount of displacement/compression of the suspension element **200**).

(33) In some cases, the thickness of the coating **150** may adjust the multi-rate spring characteristic. The thickness of the coating **150** may vary based on the desired characteristic of the spring (i.e. first spring rate, second spring rate, etc.). In an example embodiment, the thickness of the coating **150** may vary throughout the suspension element **200**. For example, the thickness of the coating **150** may be increased at the reaction portion **203** of the suspension element **200** compared to the first interface portion **201** and the second interface portion **202**.

(34) In some cases, the light emitted from the suspension element **200** may enhance specific images, logos, or text on the suspension element **200**. For example, a company logo or vehicle model may be illuminated via the light emitted. In an example embodiment, the entirety or portions of the suspension element **200** and/or the suspension assembly **100** may be made via additive manufacturing.

(35) A suspension assembly of a vehicle may be therefore provided. The suspension assembly may include a suspension element operably coupling a chassis to a wheel assembly of the vehicle, and a coating operably coupled to an exposed portion of the suspension element that is compressed or displaced responsive to a suspension event. The coating when powered may emit light to illuminate the suspension element and an area proximate to the suspension element.

(36) The suspension assembly of a vehicle of some embodiments may include additional features, modifications, augmentations and/or the like to achieve further objectives or enhance performance of the suspension system. The additional features, modifications, augmentations and/or the like may be added in any combination with each other. Below is a list of various additional features, modifications, and augmentations that can each be added individually or in any combination with each other. For example, the suspension element may be a coil spring, a leaf spring, a pneumatic spring, a stabilizer bar, a torsion spring, control arm, steering hub, suspension sleeve, suspension bellows, suspension cover, or a suspension bushing. In some cases, the light may be emitted based on an electroluminescent property of the coating. In an example embodiment, the coating may be powered by a wired connection to one or more onboard power sources of the vehicle. In some cases, the coating may be powered via a piezoelectric element. In an example embodiment, the coating may be powered via a radio frequency (RF) power element. In some cases, the suspension element may include integrated circuitry embedded within the coating. In an example embodiment, the integrated circuitry may include one or more electrical connections to other components of the vehicle, and the one or more electrical connections may be disposed at a substantially stationary portion of the suspension element. In some cases, the integrated circuitry may be formed via conductive ink painting. In an example embodiment, the integrated circuitry may include a sensor, and the sensor may provide active suspension telemetry data in real-time to a control module of the vehicle. In some cases, the sensor may be a strain gauge. In an example embodiment, an intensity of the light may vary responsive to the suspension event. In some cases, the intensity of the light may vary proportionally with the suspension event. In an example embodiment, the suspension event may be an amount of displacement or compression of the suspension element. In some cases, the suspension element may be a spring, the coating may adjust a multi-rate spring characteristic of the spring, and the multi-rate spring characteristic may be a change in spring rate as the spring is displaced. In an example embodiment, a thickness of the coating may adjust the multi-rate spring characteristic. In some cases, the coating may be an elastic coating. In an example embodiment, the coating may provide corrosion resistance for the suspension element. In some cases, the reaction portion and the exposed portion may at least partially overlap.

(37) A suspension element for operably coupling a chassis to a wheel assembly of a vehicle of an example embodiment may be provided. The suspension element may include a first interface portion in contact with a chassis component, a second interface portion in contact with a wheel assembly component, a reaction portion disposed between the first interface portion and the second interface portion to be displaced or compressed responsive to a suspension event, an exposed portion externally visible when the suspension event occurs, and a coating operably coupled to the

exposed portion of the suspension element that is displaced or compressed responsive to the suspension event. The coating when powered may emit light to illuminate the suspension element and an area proximate to the suspension element.

(38) Many modifications and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although the foregoing descriptions and the associated drawings describe exemplary embodiments in the context of certain exemplary combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the appended claims. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated as may be set forth in some of the appended claims. In cases where advantages, benefits or solutions to difficulties are described herein, it should be appreciated that such advantages, benefits and/or solutions may be applicable to some example embodiments, but not necessarily all example embodiments. Thus, any advantages, benefits or solutions described herein should not be thought of as being critical, required or essential to all embodiments or to that which is claimed herein. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A suspension assembly of a vehicle, wherein the suspension assembly comprises: a suspension element operably coupling a chassis to a wheel assembly of the vehicle; and a coating operably coupled to an exposed portion of the suspension element that is compressed or displaced responsive to a suspension event, wherein the coating when powered emits light to illuminate the suspension element and an area proximate to the suspension element.
2. The suspension assembly of claim 1, wherein the suspension element is a coil spring, a leaf spring, a pneumatic spring, a stabilizer bar, a torsion spring, control arm, steering hub, suspension sleeve, suspension bellows, suspension cover, or a suspension bushing.
3. The suspension assembly of claim 1, wherein the light is emitted based on an electroluminescent property of the coating.
4. The suspension assembly of claim 1, wherein the coating is powered by a wired connection to one or more onboard power sources of the vehicle.
5. The suspension assembly of claim 1, wherein the coating is powered via a piezoelectric element.
6. The suspension assembly of claim 1, wherein the coating is powered via a radio frequency (RF) power element.
7. The suspension assembly of claim 1, wherein the suspension element further comprises integrated circuitry embedded within the coating.
8. The suspension assembly of claim 7, wherein the integrated circuitry includes one or more electrical connections to other components of the vehicle, and wherein the one or more electrical connections are disposed at a substantially stationary portion of the suspension element.
9. The suspension assembly of claim 7, wherein the integrated circuitry is formed via conductive ink printing.
10. The suspension assembly of claim 7, wherein the integrated circuitry further comprises a sensor, wherein the sensor provides active suspension telemetry data in real-time to a control module of the vehicle.
11. The suspension assembly of claim 10, wherein the sensor is a strain gauge.
12. The suspension assembly of claim 1, wherein an intensity of the light varies responsive to the

suspension event.

13. The suspension assembly of claim 12, wherein the intensity of the light varies proportionally with the suspension event.

14. The suspension assembly of claim 13, wherein the suspension event is an amount of displacement or compression of the suspension element.

15. The suspension assembly of claim 1, wherein the suspension element is a spring, wherein the coating adjusts a multi-rate spring characteristic of the spring, and wherein the multi-rate spring characteristic is a change in spring rate as the spring is displaced.

16. The suspension assembly of claim 15, wherein a thickness of the coating adjusts the multi-rate spring characteristic.

17. The suspension assembly of claim 1, wherein the coating is an elastic coating.

18. The suspension assembly of claim 1, wherein the coating provides corrosion resistance for the suspension element.

19. A suspension element for operably coupling a chassis to a wheel assembly of a vehicle, the suspension element comprising: a first interface portion in contact with a chassis component; a second interface portion in contact with a wheel assembly component; a reaction portion disposed between the first interface portion and the second interface portion to be displaced or compressed responsive to a suspension event; an exposed portion externally visible when the suspension event occurs; and a coating operably coupled to the exposed portion of the suspension element that is displaced or compressed responsive to the suspension event, wherein the coating when powered emits light to illuminate the suspension element and an area proximate to the suspension element.

20. The suspension element of claim 19, wherein the reaction portion and the exposed portion at least partially overlap.
